H-OS: the AI Operating System of the campus

Hacktoberfest @ H-FARM · Challenge brief

In one line: build an open source **company brain** for H-FARM, a **librarian** that keeps it up to date and improves it, and **proactive agents** that act on their own and ask a human only when it matters.

The challenge

Every day H-FARM generates thousands of pieces of information: contracts, proposals, invoices, emails, meetings, projects, budgets, timesheets, CRM activities, events, guests, bookings, energy readings, decisions and commitments. Most of it lives in different applications, folders, inboxes and, ultimately, in people's heads.

What if H-FARM had one intelligence, one brain, at the center of all of this? Not another software, not another dashboard, not another database: an AI Operating System for the whole organization.

1. The Brain: one knowledge layer. A secure environment with permissioned access to H-FARM's operational knowledge. It doesn't just store information: it understands relationships, commitments, responsibilities, numbers, deadlines and expected outcomes. A librarian keeps it alive: it ingests new sources, links and deduplicates facts, and flags what is stale or contradictory.

Every person has a different AI. The intelligence is shared; what you see, your context, your objectives and your actions are personal. A CEO, a CFO, a project manager and an event manager each can see a different part of the brain.

2. The Agents: push, not pull. The system doesn't wait for questions. Proactive agents watch the brain, understand what is happening, write what they learn back into it and act to move the organization forward:

- "This proposal was sent 12 days ago and no follow-up has been recorded."
- "This invoice does not correspond to the approved proposal."
- "Revenue is on target, but the margin is deteriorating."
- "Saturday's event is at 38% of its registration target. A campaign draft is ready."
- "Building 3 is drawing power at 1 a.m. with nothing booked. I have already switched off the lights; switching off the boiler needs your OK."
- "The guest arriving at 3 p.m. has already met two of us: here is the brief."

Management by exception. EVENT → CONTEXT → REASONING → ACTION. Something happens, the system understands what it means, identifies who should care, checks it against rules and history, then alerts, proposes or, where authorized, acts. Instead of "Show me everything", it answers "Show me what needs my attention, the rest is already done". And it learns: every project and every decision makes it smarter.

The hack challenge

How would you architect the first working version of H-OS: a company brain, and the proactive agents that act on it? For the Brain: the knowledge layer, the organizational graph, who sees what. For the Librarian: how the brain stays current and gets better over time. For the Agents: how LLMs, decision systems, rules and agents work together, and the event and action engine. Around both: the interfaces (for people, and for other agents through APIs or MCP), the infrastructure (open-weight models on hardware we own are welcome) and governance.

Access is one of the hard parts. Beyond retrieving the right data and bringing it into one place, someone has to decide who sees what, and keep it right as people and roles change. You can start from one person with full access who delegates: "I want the head of the Business School to see this, this and this."

The prototype

Pick one of these two workflows, or propose your own:

- **The life of a campus event:** Event created → Room booked → Registrations → Guest check-in → Energy used → Follow-up
- **From contract to cash:** Deal signed → Order → Delivery → Invoice → Payment

You can build a small synthetic dataset, or start from public sources such as the events page on h-farm.com. You could give at least three roles different access, then show your agents spotting what is missing, inconsistent or late, and acting on it.

You don't have to start from zero. For the Brain and the Librarian (company brains, LLM wikis): Karpathy's LLM wiki, PipesHub, Cognee. For the Agents (agent-first): QM by YC, Open Instinct, Paperclip.

What you should deliver

- A public GitHub repo under an open source license, submitted on the Hacktoberfest platform, that runs from its README in minutes.

- The repo contains three parts: the Brain; the Librarian that maintains and improves it; and at least one agent or a multi-agent team that, on a concrete case, acts on its own and asks permission when needed (e.g. lights off by itself, boiler only with an OK).
- A working loop (e.g. EVENT → CONTEXT → REASONING → ACTION) with at least three roles and every answer citing its sources.
- A one-page architecture: what is real, what is mocked.
- A live demo.

Guardrails

- Synthetic or public data only: no real personal data, no data about minors, no access to H-FARM systems beyond public information.
- Anything that leaves the organization, spends money, writes to a system of record or changes permissions should be carefully evaluated for a human approval step.
- No invoices issued, no accounting entries: the system reconciles and flags.
- Nothing that monitors or scores individuals.

The north star

Today people operate software. Tomorrow people express intentions, and AI operates the organization.

Can we build an organization that understands itself, and helps every person inside it perform better?